L1.5 Differentiation Formulae

§2.3 — what you need to know

The five rules

① **constant** $\dfrac{d}{dx}(c) = 0$

② **power** $\dfrac{d}{dx}\left(x^n\right) = nx^{n-1}$ every real n — negative and fractional too

③ **sum** $\dfrac{d}{dx}\left(f \pm g\right) = \dfrac{df}{dx} \pm \dfrac{dg}{dx}$ $\dfrac{d}{dx}\left(cf\right) = c\,\dfrac{df}{dx}$

THE TWO THAT ARE NOT TERM BY TERM

④ **product** $\dfrac{d}{dx}\big(f(x)g(x)\big) = f(x)g'(x) + g(x)f'(x)$

⑤ **quotient** $\dfrac{d}{dx}\left(\dfrac{f(x)}{g(x)}\right) = \dfrac{gf' - fg'}{g^2}$

"low d high, minus high d low, over the square of what's below"

Each function is **kept once and differentiated once**. The quotient's numerator is a subtraction, so its order is not free — swapping the two products gives the negative of the right answer.

REWRITE AS A POWER FIRST

The power rule applies to x^n and nothing else, so roots and reciprocals are rewritten *before* they are differentiated. **This is the step that gets skipped.**

$\sqrt{x} \to x^{1/2}$ $\dfrac{1}{x^3} \to x^{-3}$ $\sqrt[3]{x^2} \to x^{2/3}$

And you can **always take out the** c: $\dfrac{d}{dx}\left(20x^{1/2}\right) = 20\,\dfrac{d}{dx}\left(x^{1/2}\right) = 20 \cdot \tfrac{1}{2}x^{-1/2} = \dfrac{10}{\sqrt{x}}$

WHERE nx^{n-1} COMES FROM

Straight out of the definition. The coefficients of $(x+h)^3$ are row 3 of **Pascal's triangle** — 1 3 3 1 — and every term then has one of **three fates**:

$$\frac{(x+h)^3 - x^3}{h} = \frac{\overbrace{x^3 - x^3}^{\text{cancels}} + \overbrace{3x^2h}^{h \text{ divides out}} + \overbrace{3xh^2 + h^3}^{\to 0}}{h} = 3x^2 + 3xh + h^2 \xrightarrow[h\to 0]{} 3x^2$$

The coefficient is the second entry of row n, which is always n; the exponent is always one less. That is the whole rule — and it is why it works for every n.

SECOND DERIVATIVES AND HORIZONTAL TANGENTS

$f''(x) = \dfrac{d^2y}{dx^2}$ is the same five rules **run twice**. Nothing new.

A **horizontal tangent** is a slope of 0, so you find one by **solving** $f'(x) = 0$. For $y = x^3 - x$: $3x^2 - 1 = 0 \Rightarrow \boxed{x = \pm\frac{1}{\sqrt{3}}}$. *This is the first time the derivative is used to* ***find*** *something instead of to report it.*

`M` (max or min) on f → a **zero** on f' → —

`IP` on f → an `M` on f' → a **zero** on f''

WATCH OUT

✗ ~~the derivative of a product is the product of the derivatives~~ → **it is** $fg' + gf'$ **— expanding** $(x^2 - 1)(x + 4)$ **proves it**

✗ ~~the quotient's numerator is $fg' - gf'$~~ → **it is** $gf' - fg'$ **— low d high** ***minus*** **high d low**

✗ ~~$\sqrt{x}$ needs its own rule~~ → **rewrite it as** $x^{1/2}$ **and it** ***is*** **the power rule**

✗ ~~f'' needs a new technique~~ → **it is the same rules, run twice**

✗ ~~the power rule works on 2^x~~ → **the** ***exponent*** **has to be the constant, not the base**

✗ ~~every quotient needs the quotient rule~~ → **if the bottom is a single power, split the fraction instead**